# Francesca Giardine

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## Education

**Ph.D.** **University of California, Berkeley** *May 2026*
Doctor of Philosophy in Electrical Engineering and Computer Sciences
Dissertation: "Analysis and Operation of Flying Capacitor Multilevel Converters: Single-Phase Applications and Fundamental Electromagnetic Interference Scaling"
Advisor: Dr. Robert Pilawa-Podgurski

**B.S.** **Smith College** *May 2020*
Bachelor of Science in Engineering Science & Mathematics and Statistics
*Summa Cum Laude*

## Honors and Awards

### Fellowships

National Science Foundation Graduate Research Fellowship *Sept. 2022 - Sept. 2025*
- Tuition and stipend for 3 years of graduate study in STEM.

Hearts to Humanity Eternal Graduate Research Grant *Sept. 2023 - Sept. 2024*
- $10,000 grant for independent research.

Berkeley EECS Department Fellowship *Aug. 2020*
- Semester-long fellowship for select incoming students.

### Recognitions

Rising Stars in EECS *October 2025*
- Selective 2-day program organized by MIT and Boston University to provide preparation for junior faculty positions.

WiscProf Future Faculty Program Participant *May 2025*
- Selective 4-day program sponsored by the University of Wisconsin, Madison providing training for faculty positions in academia.

Best Paper Award, ECCE Europe 2024 *September 2024*
- Awarded to 10 of 300 papers presented at ECCE Europe for paper [C4].

NextProf Nexus Future Faculty Program Participant *August 2024*
- Selective program sponsored by the University of Michigan, UC Berkeley, and Georgia Tech giving participants the opportunity to explore and prepare for a faculty position in academia.

Adeline Devor Penberthy, Class of 1945, Memorial Prize *May 2020*
- Awarded to an undergraduate engineering major for academic excellence in engineering and outstanding contributions toward building a community of learners within the Smith College Picker Engineering Program.

## Research Experience

**Postdoctoral Research Associate in Stillwell Power Group** *Aug. 2026 - Present*
*University of Illinois Urbana-Champaign* *Urbana, IL*

- Developing electromagnetic interference (EMI) test setup for the power research area, with an emphasis on long-term usability.
- Analyzing EMI in power converters for electrified transportation.
- Mentoring student research projects, specifically in EMI and capacitor-based converters.

**Pilawa Research Group** *Aug. 2020 - May 2026*
*University of California, Berkeley* *Berkeley, CA*

- Electromagnetic interference study for the flying capacitor multilevel converter.
- Formulated control technique to improve efficiency and electromagnetic interference of flying capacitor multilevel inverters.
- Simulated and designed advanced hybrid switched-capacitor converters for use in DC-AC grid-connected systems.

**Power Electronic Device Characterization Internship** *June 2023 - Aug. 2023*
*National Laboratory of the Rockies* *Golden, CO*

- Designed a test setup for switch characterization for power electronics.
- Performed simulation, printed circuit board layout, and investigated advanced sensing techniques to achieve high-bandwidth performance.

**Researcher in the Renewable and Appropriate Energy Lab** *July 2020 - Aug. 2020*
*University of California, Berkeley* *Berkeley, CA*

- Extended research beginning in the SUPERB REU under Professor Daniel Kammen in the Energy Resources Group.
- Performed statistical and spatial analyses in Python and GIS to isolate candidate communities for microgrid development

**SUPERB Research Experience for Undergraduates** *June 2019 - Aug. 2019*
*University of California, Berkeley* *Berkeley, CA*

- Analyzed geospatial data in RStudio and GIS to classify community types in California with emphasis on environmental predictors.
- Selected as the sole student from the SUPERB REU to present research at NSF REU Research Symposium in Alexandria, VA, in October 2019.

**Capstone to Work Research Team** *Sept. 2018 - May 2020*
*Smith College* *Northampton, MA*

- Reviewed qualitative and quantitative data from a large pool of first year engineering employees to analyze the impacts of the engineering capstone experiences on employment after college.

## Publications

### Refereed Journal Papers

**[J7]** S. Krishnan, **F. Giardine**, T. Mahbub, L. Horowitz, and R. C. N. Pilawa-Podgurski, "A High-Performance Flying Capacitor Multilevel Dual Active Bridge DC-DC Converter for Isolated Automotive Systems," *IEEE Open Journal of Power Electronics*, vol. 7, pp. 2701–2713, 2026. DOI: 10.1109/OJPEL.2026.3733626

**[J6]** **F. Giardine**, S. Krishnan, Y. Wu, L. Horowitz, and R. C. N. Pilawa-Podgurski, "A Variable Frequency Technique for EMI and Efficiency Improvements in High-Level Count Flying Capacitor Multilevel Converters," *IEEE Open Journal of Power Electronics*, pp. 267–280, 2025. DOI: 10.1109/OJPEL.2025.3646981

[**J5**] N. C. Brooks, **F. Giardine**, and R. C. N. Pilawa-Podgurski, "DC-Link Capacitors for Twice-Line Frequency Power Decoupling: Design-Oriented Figures-of-Merit With Empirical Application," *IEEE Transactions on Power Electronics*, vol. 39, no. 6, pp. 6569–6573, 2024. DOI: 10.1109/TPEL.2023.3305153

[**J4**] J. Ford, M. Paretti, D. Kotys-Schwartz, S. Howe, C. Gerwitz, J. Deters, T. Mahmud, R. Ott, N. Alvarez, D. Knight, C. Hernandez, L. Rosenbauer, A. Kary, and **F. Giardine**, "Transitioning from Capstone Design Courses to Workplaces: A Study of New Engineers' First Three Months," *International Journal of Engineering Education*, 2019.

## Journal Papers (In Progress)

[**J3**] **F. Giardine**, K. Fernandez, and R. C. N. Pilawa-Podgurski, "A Two-Stage Non-Isolated Hybrid Switched-Capacitor Microinverter Utilizing a Multi-Resonant DC-DC Stage with Startup Functionality," *IEEE Open Journal of Power Electronics*, [Conditionally Accepted].

[**J2**] **F. Giardine**, S. Krishnan, M. Nerenberg*, and R. C. N. Pilawa-Podgurski, "On the Scaling of Common-Mode EMI in the Flying Capacitor Multilevel Converter," [In Revision].

[**J1**] E. Krause, **F. Giardine**, N. Biesterfeld, and R. C. N. Pilawa-Podgurski, "Analysis of Steady-State Balancing in the Flying Capacitor Multilevel Converter Considering Capacitor Voltage Ripple," [In Revision].

## Refereed Conference Papers

[**C11**] Y. Zhao*, J. Armstrong*, **F. Giardine**, and R. C. N. Pilawa-Podgurski, "A High-Performance 6-Level Flying Capacitor Multilevel Totem-Pole Bidirectional DC-AC Converter for Single-Phase Systems," in *2026 IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2026, pp. 71–77. DOI: 10.1109/APEC51134.2026.11516908

[**C10**] S. Krishnan, **F. Giardine**, S. T. Mahbub, L. Horowitz, and R. C. N. Pilawa-Podgurski, "A High-Performance Flying Capacitor Multilevel Dual Active Bridge DC-DC Converter for Isolated Automotive Systems," in *2026 IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2026, pp. 3621–3627. DOI: 10.1109/APEC51134.2026.11516761

[**C9**] **F. Giardine**, S. Krishnan, M. Nerenberg*, and R. C. N. Pilawa-Podgurski, "An Analytical Model for Common-Mode EMI in Flying Capacitor Multilevel Converters," in *2025 IEEE 26th Workshop on Control and Modeling for Power Electronics (COMPEL)*, 2025, pp. 1–7. DOI: 10.1109/COMPEL57166.2025.11121281

[**C8**] E. Krause, N. Biesterfeld, **F. Giardine**, and R. C. N. Pilawa-Podgurski, "Analysis of Steady-State Balancing in the Flying Capacitor Multilevel Converter Considering Capacitor Voltage Ripple," in *2025 IEEE 26th Workshop on Control and Modeling for Power Electronics (COMPEL)*, 2025, pp. 1–8. DOI: 10.1109/COMPEL57166.2025.11121246

[**C7**] S. Krishnan, **F. Giardine**, J. Zou, R. A. Abramson, and R. C. N. Pilawa-Podgurski, "Design and Implementation of a GaN-Based, Cascaded Isolated DC-DC Converter for Satellite Applications," in *2025 IEEE/AIAA Transportation Electrification Conference and Electric Aircraft Technologies Symposium (ITEC+EATS)*, 2025, pp. 1–6. DOI: 10.1109/ITEC63604.2025.11097988

[**C6**] **F. Giardine**, S. Krishnan, L. Horowitz, and R. C. N. Pilawa-Podgurski, "A Variable Frequency Technique for EMI and Efficiency Improvements in High-Level Count Flying Capacitor Multilevel Converters," in *2025 IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2025, pp. 151–156. DOI: 10.1109/APEC48143.2025.10977102

[**C5**] **F. Giardine**, Y. Wu, and R. C. N. Pilawa-Podgurski, "A Variable Switching Frequency Control Technique for DC-AC Flying Capacitor Multilevel Converters to Improve Efficiency and Inductor Utilization," in *2024 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2024, pp. 3473–3478. DOI: 10.1109/ECCE55643.2024.10861273

[**C4**] **F. Giardine**, K. Fernandez, and R. C. N. Pilawa-Podgurski, "A Two-Stage Non-Isolated Hybrid Switched-Capacitor Microinverter Utilizing a Fixed-Ratio Resonant DC-DC Stage with Startup Functionality and Flying Capacitor Multilevel DC-AC Stage," in *2024 Energy Conversion Congress & Expo Europe (ECCE Europe)*, [Best Paper Award], 2024, pp. 1–7. DOI: 10.1109/ECCEEurope62508.2024.10752033

[**C3**] **F. Giardine**, N. C. Brooks, K. Fernandez, and R. C. Pilawa-Podgurski, "Utilizing Harmonic Injection to Reduce Energy Storage and Required Capacitance in an Active Series-Stacked Energy Buffer for Single-Phase Systems," in *2022 IEEE 23rd Workshop on Control and Modeling for Power Electronics (COMPEL)*, 2022, pp. 1–7. DOI: 10.1109/COMPEL53829.2022.9829980

[**C2**] C. Gerwitz, **F. Giardine**, R. Ott, and A. Kary, "Women's Unique Challenges in the Transitions to Engineering Work," in *2020 ASEE Annual Conference & Exposition*, 2020.

[**C1**] S. Howe, M. Paretti, R. Ott, J. Deters, C. Gerwitz, **F. Giardine**, C. Hernandez, and A. Kary, "Women's Experiences in the Transition from Capstone Design Courses to Engineering Workplaces," in *2019 ASEE Annual Conference & Exposition*, 2019. DOI: 20.500.12592/23pjcnp

*denotes undergraduate student

## Mentoring

**UIUC**

Jay Lee (Undergraduate) — *September 2026 - Present*
*Ripple-Free Multilevel Power Conversion*

Abhinav Srinivasan (Undergraduate) — *September 2026 - Present*
*800 V Power Conversion for Datacenters*

Charlie Gunderson (Undergraduate) — *September 2026 - Present*
*Custom Coupled-Inductor Design*

**UC Berkeley**

Jack Armstrong (Undergraduate/Graduate) *May 2025 - Present*
*Rectifier Control for a High-Performance Totem-Pole PFC Converter*
*High Performance Active Capacitor for DC-Link Ripple Reduction*

Stephanie Schaupp (Undergraduate) *September 2025 - May 2026*
*High Voltage Safety Enclosures*

Yayun Zhao (Undergraduate, now pursuing Ph.D at UT Austin) *October 2024 - March 2026*
*Design of a High-Performance Totem-Pole PFC Converter*

Nikhil Ograin (Undergraduate, now at Apple) *September 2024 - May 2025*
*A Resonant Flying Capacitor Multilevel Inverter*

## Service and Leadership

Berkeley EECS Graduate Admissions Reader *2023, 2024*

IEEE Power and Energy Society *Jan. 2021 - December 2025*

- 2021: Member-at-Large
- 2022, 2024: Vice Chair
- 2023: Chair

EECS Graduate Student Association Committee Member *Sept. 2021 - Sept. 2022*